Data Analysis
AMS315

Chapter 7

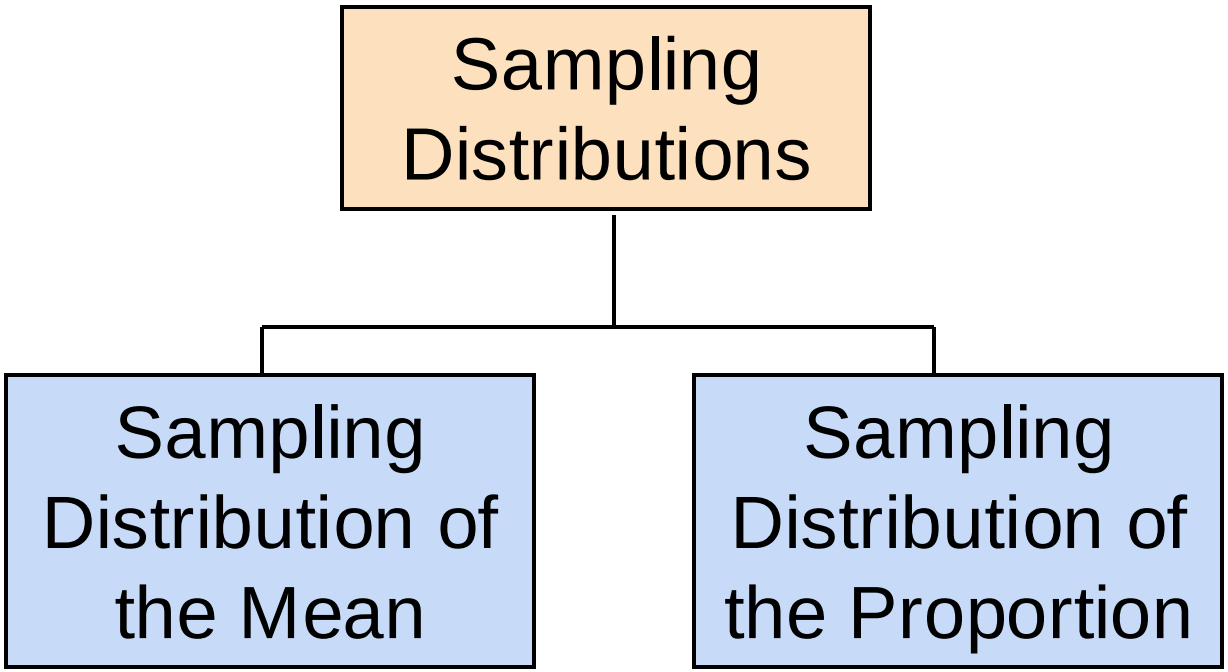
Sampling Distributions

Learning Objectives

In this chapter, you learn:

- The concept of the sampling distribution
- To compute probabilities related to the sample mean and the sample proportion
- The importance of the Central Limit Theorem
- To distinguish between different survey sampling methods
- To evaluate survey worthiness and survey errors

Sampling Distributions



Sampling Distributions

- A sampling distribution is a distribution of all of the possible values of a statistic for a given size sample selected from a population

Developing a Sampling Distribution

- **Assume there is a population …**
- Population size N=4
- Random variable, X, is age of individuals
- Values of X: 18, 20, 22, 24 (years)

Developing a Sampling Distribution

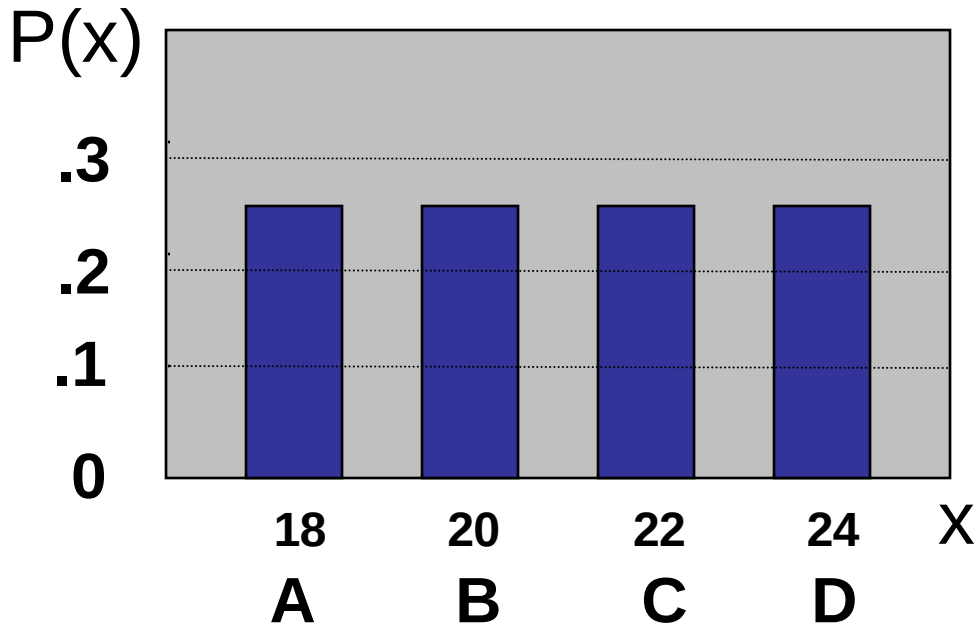
(continued)

Summary Measures for the Population Distribution:

$$\mu = \frac{\sum X_i}{N}$$

$$= \frac{18 + 20 + 22 + 24}{4} = 21$$

$$\sigma = \sqrt{\frac{\sum (X_i - \mu)^2}{N}} = 2.236$$



Uniform Distribution

Developing a Sampling Distribution

(continued)

Now consider all possible samples of size n=2

1st Obs	2nd Observation			
	18	20	22	24
18	18,18	18,20	18,22	18,24
20	20,18	20,20	20,22	20,24
22	22,18	22,20	22,22	22,24
24	24,18	24,20	24,22	24,24

16 possible samples (sampling with replacement)

16 Sample Means

1st Obs	2nd Observation			
	18	**20**	**22**	**24**
18	18	19	20	21
20	19	20	21	22
22	20	21	22	23
24	21	22	23	24

Developing a Sampling Distribution

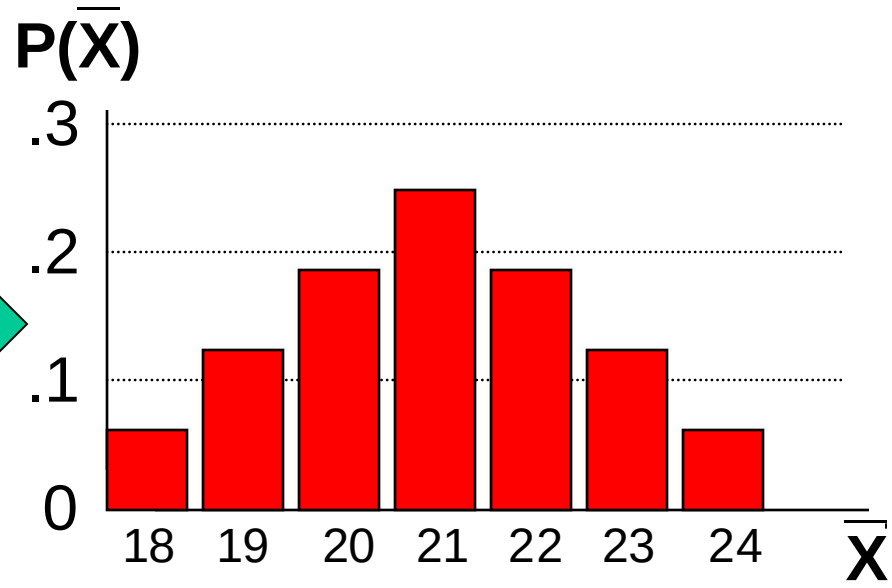
(continued)

Sampling Distribution of All Sample Means

16 Sample Means

1st Obs	2nd Observation			
	18	**20**	**22**	**24**
18	18	19	20	21
20	19	20	21	22
22	20	21	22	23
24	21	22	23	24

Sample Means Distribution



(no longer uniform)

Developing a Sampling Distribution

(continued)

Summary Measures of this Sampling Distribution:

$$\mu_{\bar{X}} = \frac{\sum \bar{X}_i}{N} = \frac{18 + 19 + 21 + \cdots + 24}{16} = 21$$

$$\sigma_{\bar{X}} = \sqrt{\frac{\sum (\bar{X}_i - \mu_{\bar{X}})^2}{N}}$$

$$= \sqrt{\frac{(18 - 21)^2 + (19 - 21)^2 + \cdots + (24 - 21)^2}{16}} = 1.58$$

Comparing the Population with its Sampling Distribution

Population

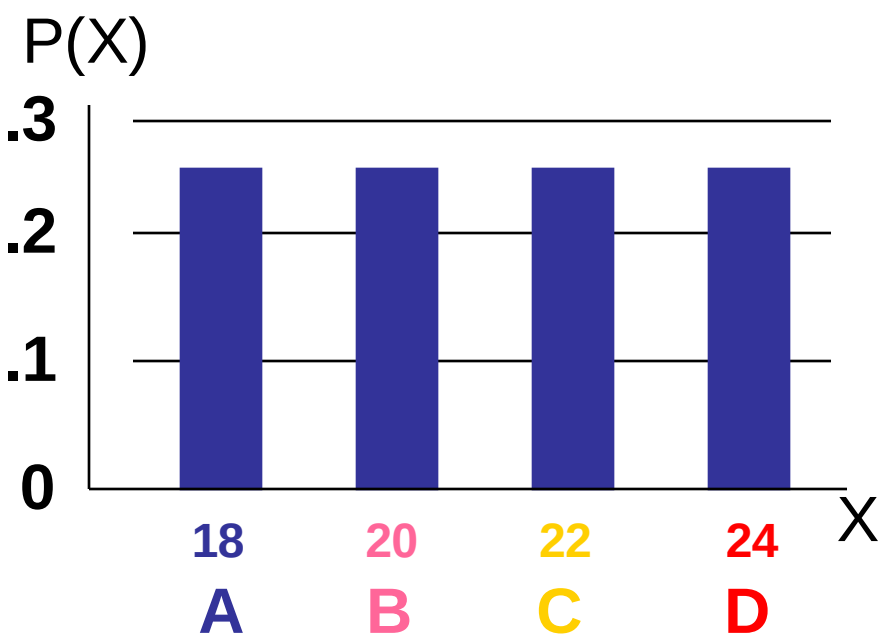
$N = 4$

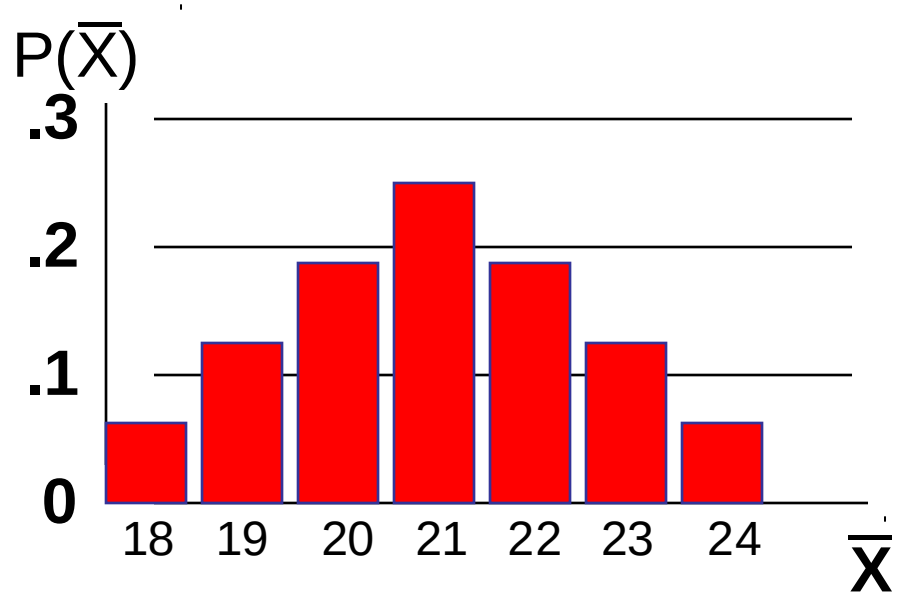
$\mu = 21 \quad \sigma = 2.236$

Sample Means Distribution

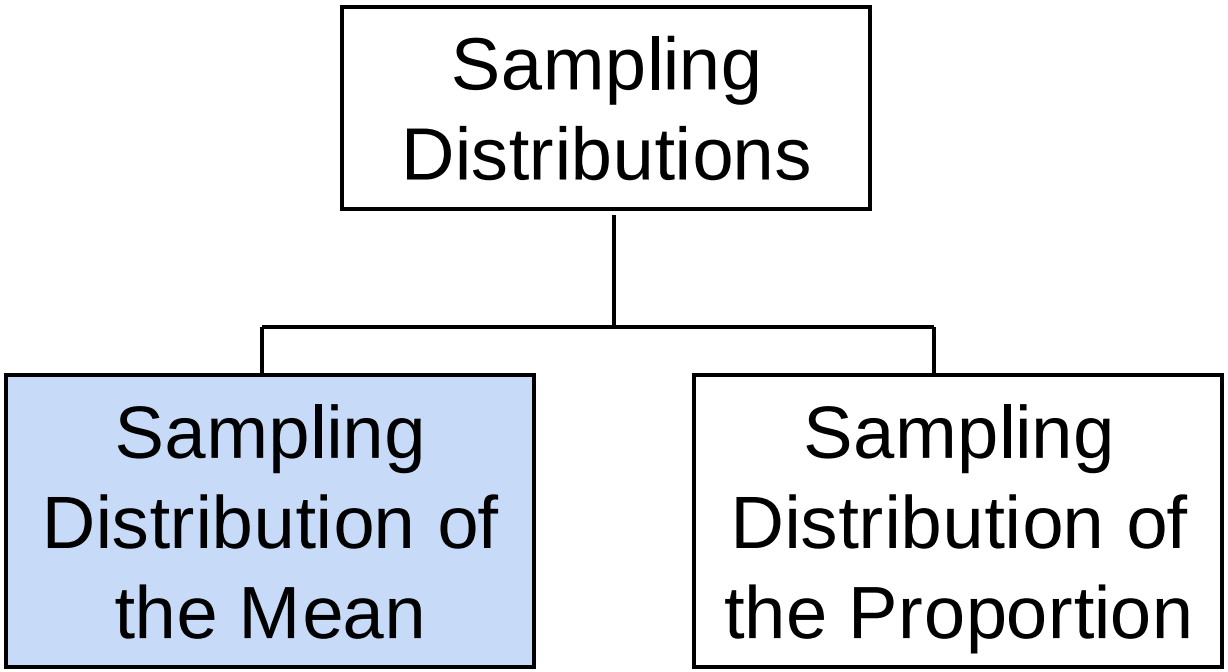
$n = 2$

$\mu_{\bar{X}} = 21 \quad \sigma_{\bar{X}} = 1.58$





Sampling Distribution of the Mean



Standard Error of the Mean

- Different samples of the same size from the same population will yield different sample means
- A measure of the variability in the mean from sample to sample is given by the Standard Error of the Mean:

  (This assumes that sampling is with replacement or sampling is without replacement from an infinite population)

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

- Note that the standard error of the mean decreases as the sample size increases

If the Population is Normal

- If a population is normal with mean μ and standard deviation σ, the sampling distribution of $\bar{X}$ is also normally distributed with

$$\boxed{\mu_{\bar{X}} = \mu}$$

and

$$\boxed{\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}}$$

Z-value for Sampling Distribution of the Mean

- Z-value for the sampling distribution of $\bar{X}$:

$$Z = \frac{(\bar{X} - \mu_{\bar{X}})}{\sigma_{\bar{X}}} = \frac{(\bar{X} - \mu)}{\frac{\sigma}{\sqrt{n}}}$$

where: $\bar{X}$ = sample mean

μ = population mean

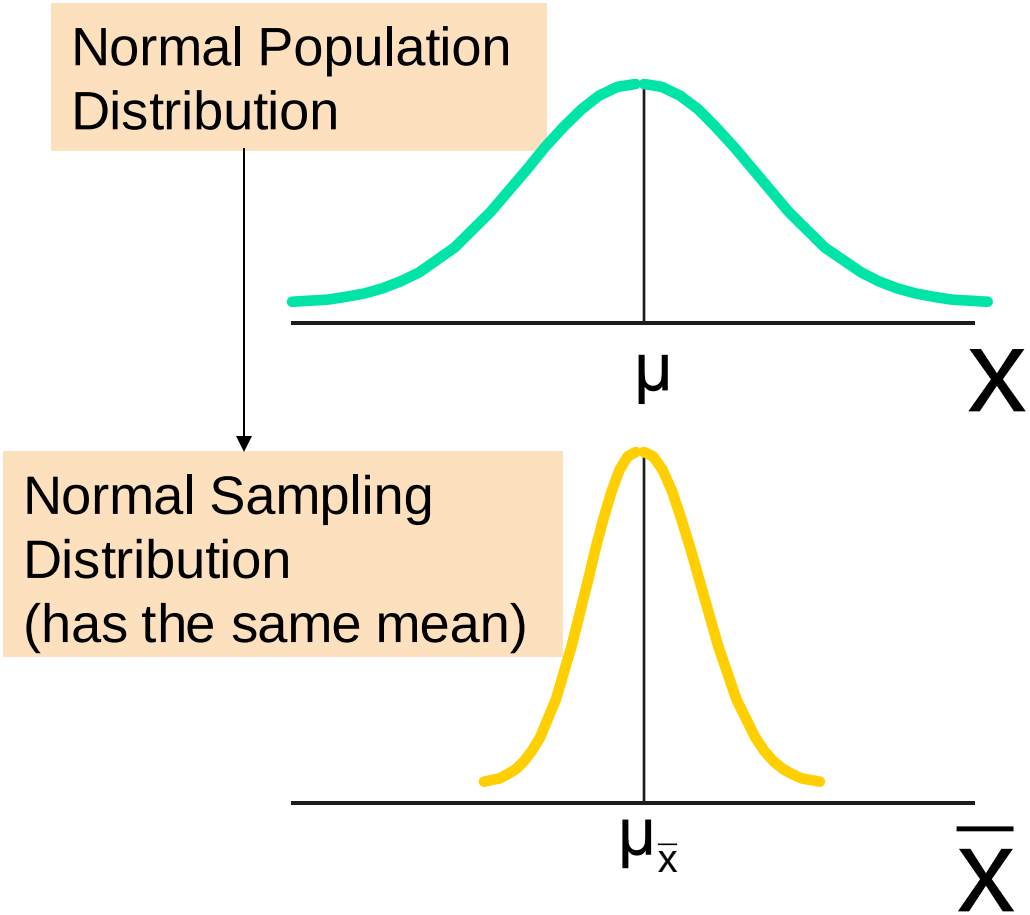
σ = population standard deviation

n = sample size

Sampling Distribution Properties

- $$\mu_{\bar{x}} = \mu$$

(i.e. $\bar{x}$ is unbiased)

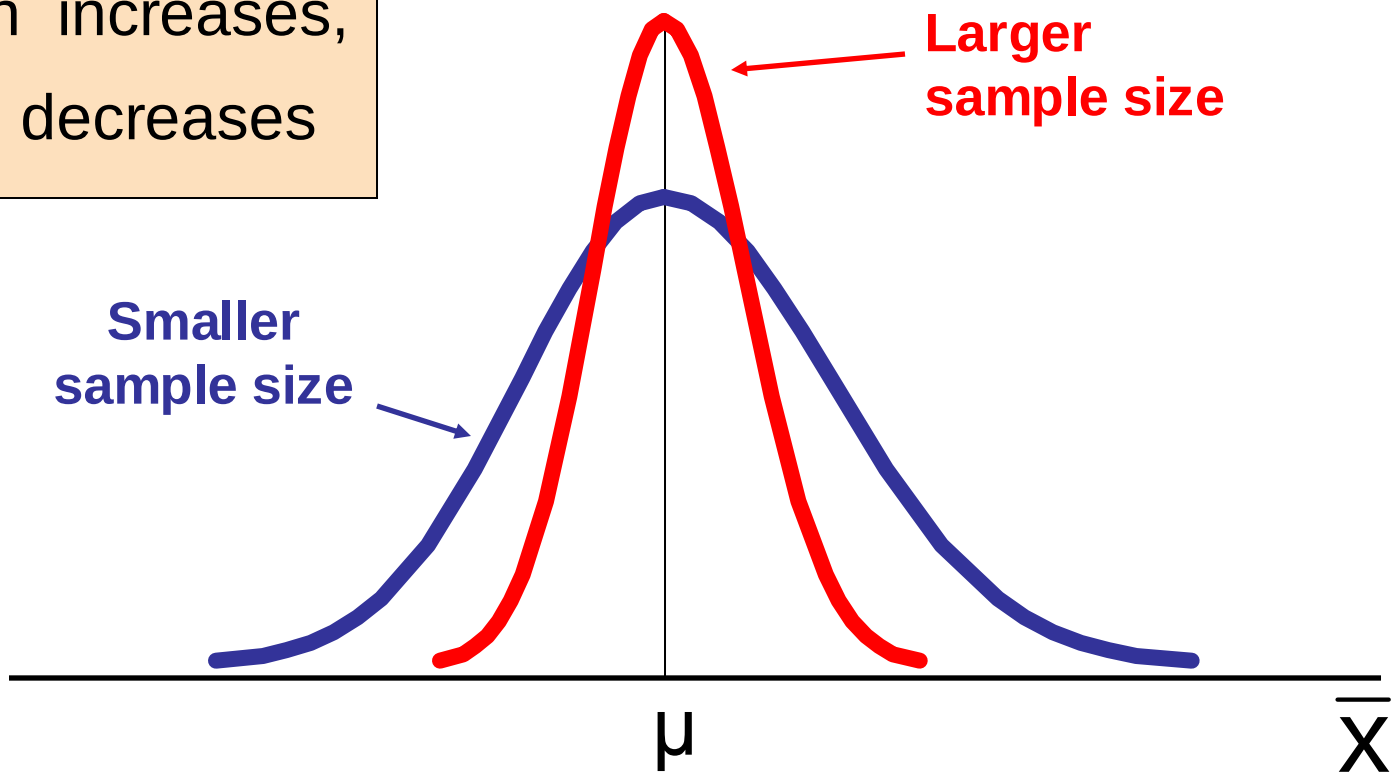


Sampling Distribution Properties

(continued)

As n increases,

$\sigma_{\bar{x}}$ decreases



If the Population is **not** Normal

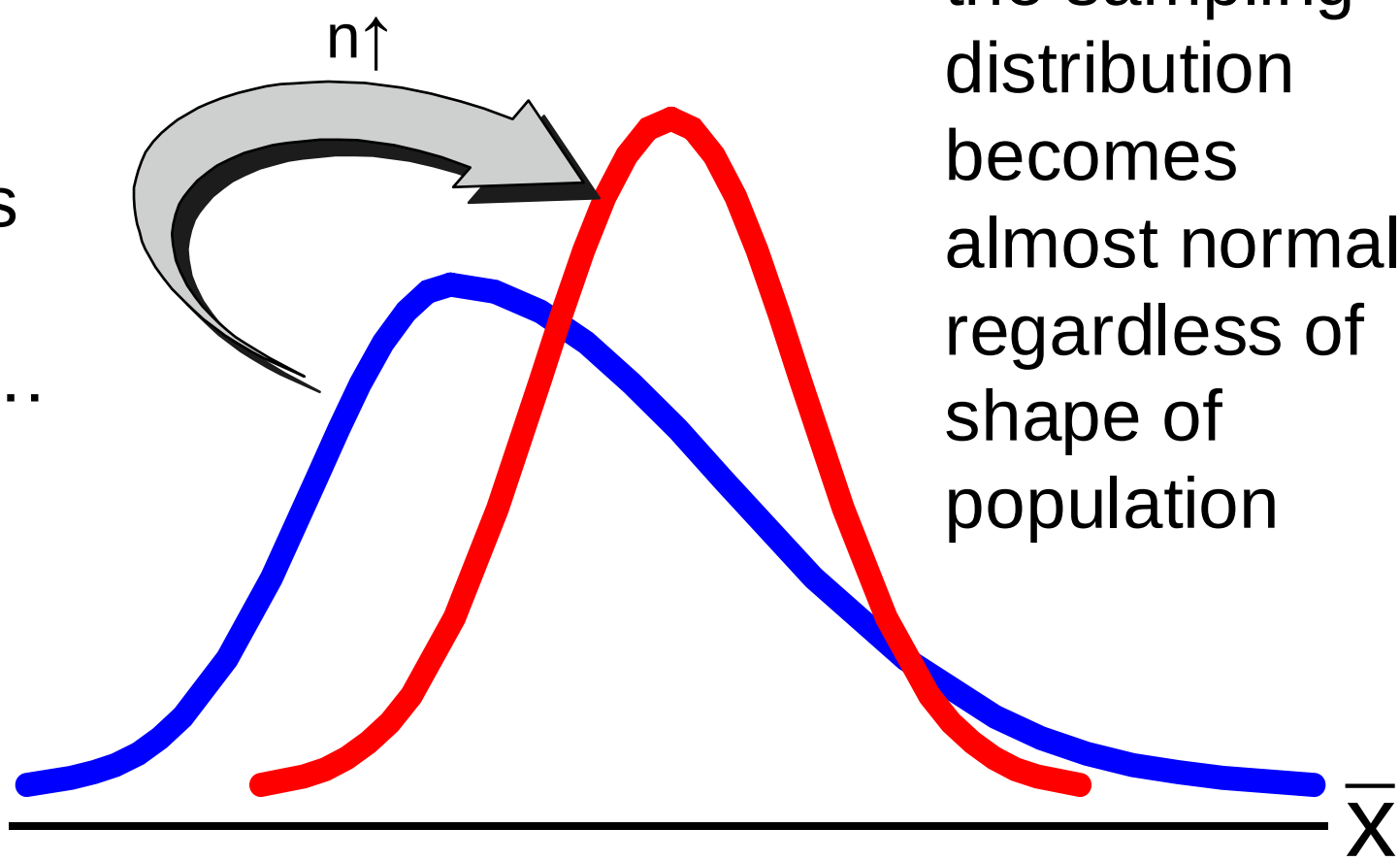
- We can apply the Central Limit Theorem:
  - Even if the population is not normal,
  - sample means from the population will be approximately normal as long as the sample size is large enough (n>30).

Properties of the sampling distribution:

$$\boxed{\mu_{\bar{x}} = \mu} \quad \text{and} \quad \boxed{\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}}$$

Central Limit Theorem

As the sample size gets large enough…



the sampling distribution becomes almost normal regardless of shape of population

Central Limit Theorem

- The central limit theorem says that the sampling distribution of the mean will always be normally distributed, as long as the sample size is large enough (n>30).

- Regardless of whether the population has a normal, Poisson, binomial, or any other distribution, the sampling distribution of the mean will be normal.

If the Population is **not** Normal

(continued)

Sampling distribution properties:

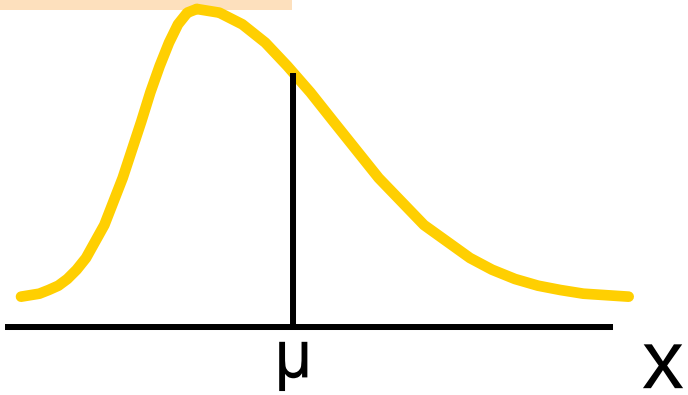
Central Tendency

$$\mu_{\bar{x}} = \mu$$

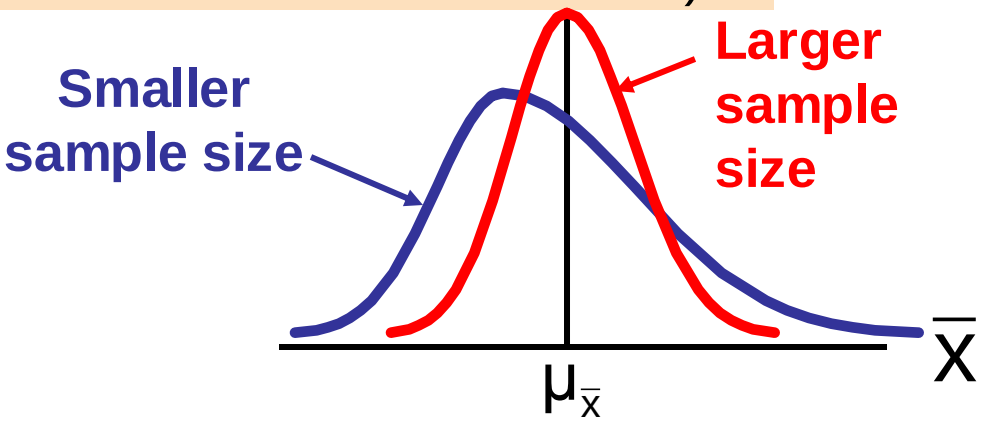
Variation

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Population Distribution



Sampling Distribution
(becomes normal as n increases)



How Large is Large Enough?

- For most distributions, $n > 30$ will give a sampling distribution that is nearly normal
- For fairly symmetric distributions, $n > 15$
- For normal population distributions, the sampling distribution of the mean is always normally distributed

Example

- Suppose a population has mean $\mu = 8$ and standard deviation $\sigma = 3$. Suppose a random sample of size $n = 36$ is selected.

- What is the probability that the sample mean is between 7.8 and 8.2?

Example

(continued)

Solution:

- Even if the population is not normally distributed, the central limit theorem can be used (n > 30)
- So, the sampling distribution of $\bar{X}$ is approximately normal,
  - with mean $\mu_{\bar{x}} = 8$
  - and standard deviation $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{3}{\sqrt{36}} = 0.5$

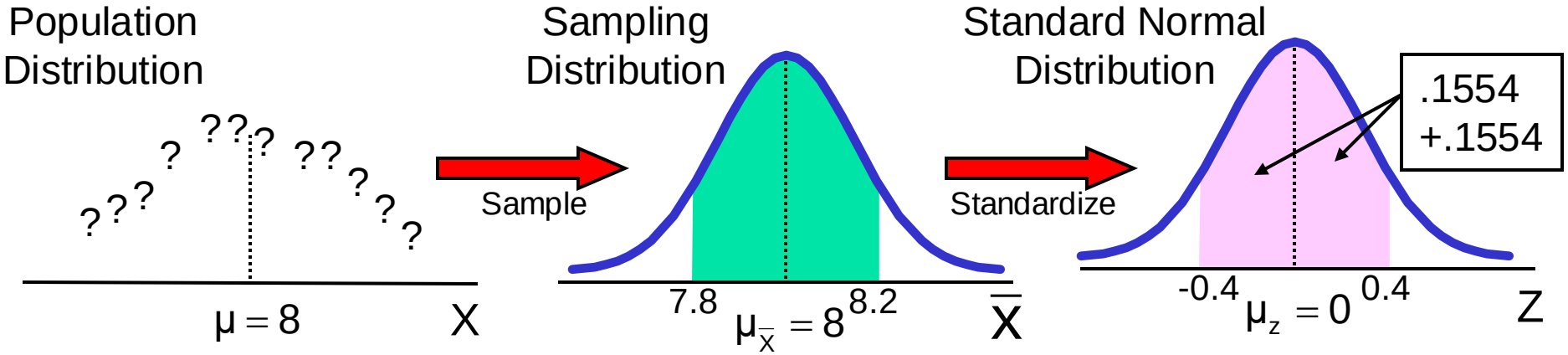
Example

(continued)

Solution (continued):

$$P(7.8 < \bar{X} < 8.2) = P\left(\frac{7.8-8}{3/\sqrt{36}} < \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} < \frac{8.2-8}{3/\sqrt{36}}\right)$$

$$= P(-0.4 < Z < 0.4) = \boxed{0.3108}$$



Class Tutorial

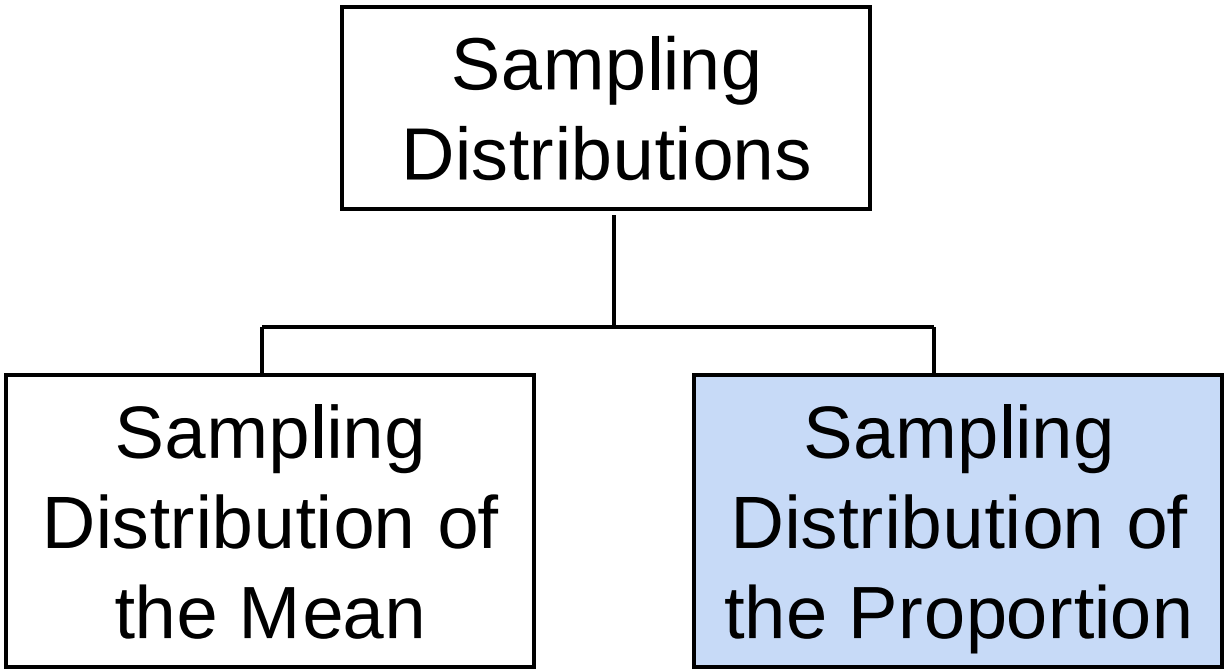
- A sample size of 100 values is randomly selected from a population with mean 85 and standard deviation of 5.25. Assume the random values *x* has normal distribution.

- Find $P(x < 75)$
- Find $P(\bar{x} < 75)$

Class Tutorial

- A sample of 100 measurements are taken from a population whose value of μ is assumed to be 98.6.
- If the sample standard deviation s = 1.5, find $P(\overline{x} \geq 99)$.

Sampling Distribution of the Proportion



Population Proportions

π = the proportion of the population having some characteristic

- Sample proportion (p) provides an estimate of π:

$$p = \frac{X}{n} = \frac{\text{number of items in the sample having the characteristic of interest}}{\text{sample size}}$$

- $0 \leq p \leq 1$
- p has a binomial distribution

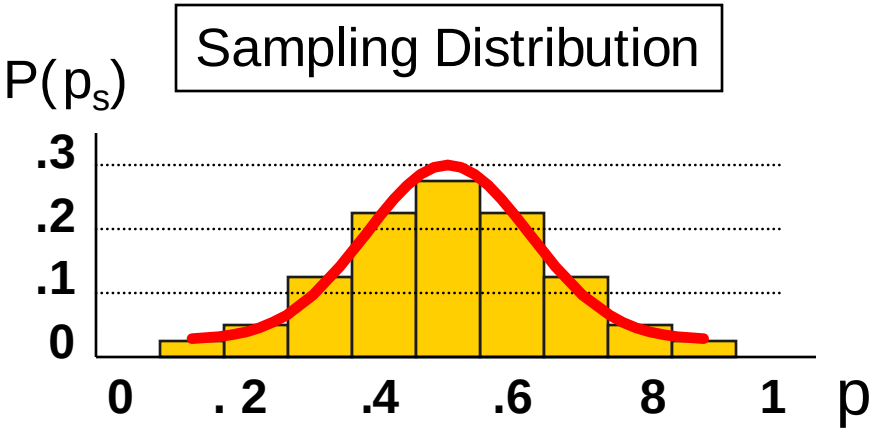
(assuming sampling with replacement from a finite population or without replacement from an infinite population)

Sampling Distribution of p

- Approximated by a normal distribution if:
  - $np \geq 5$

    and

    $n(1-p) \geq 5$



where

$$\mu_p = \pi$$

and

$$\sigma_p = \sqrt{\frac{\pi(1-\pi)}{n}}$$

(where π = population proportion)

Z-Value for Proportions

Standardize p to a Z value with the formula:

$$Z = \frac{p - \pi}{\sigma_p} = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$$

Example

- If the true proportion of voters who support Proposition A is $\pi = 0.4$, what is the probability that a sample of size 200 yields a sample proportion between 0.40 and 0.45?

- i.e.: **if $\pi = 0.4$ and $n = 200$, what is $P(0.40 \le p \le 0.45)$?**

Example

(continued)

- **if $\pi = 0.4$ and $n = 200$, what is $P(0.40 \le p \le 0.45)$?**

Find σ_p:

$$\sigma_p = \sqrt{\frac{\pi(1-\pi)}{n}} = \sqrt{\frac{0.4(1-0.4)}{200}} = 0.03464$$

Convert to standard normal:

$$P(0.40 \le p \le 0.45) = P\left(\frac{0.40-0.40}{0.03464} \le Z \le \frac{0.45-0.40}{0.03464}\right)$$

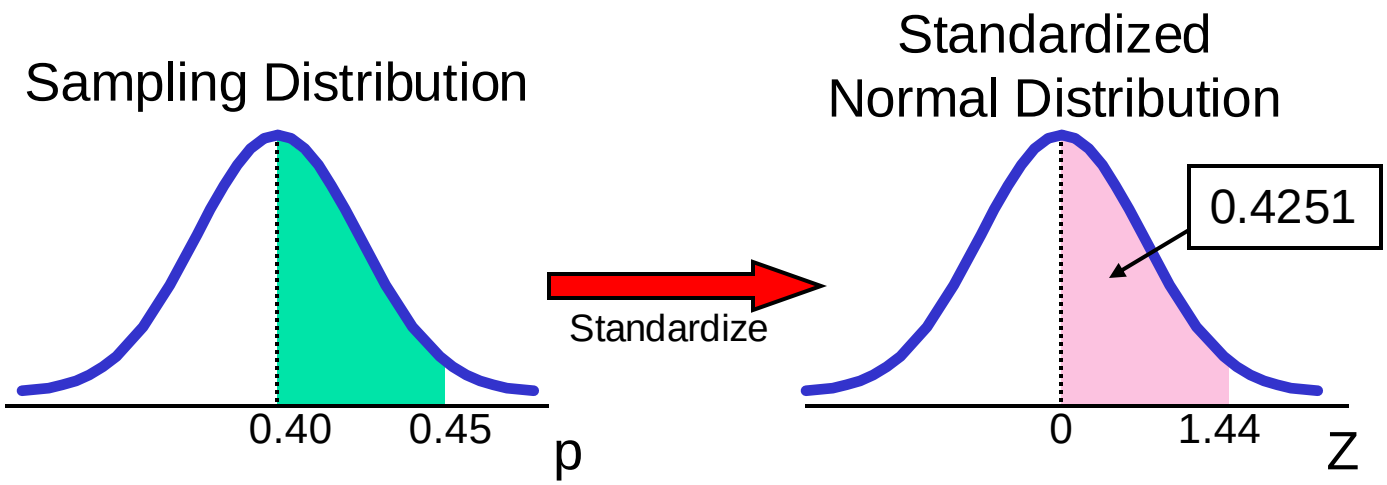
$$= P(0 \le Z \le 1.44)$$

Example

(continued)

- **if $\pi = 0.4$ and $n = 200$, what is $P(0.40 \le p \le 0.45)$?**

Use standard normal table: $P(0 \le Z \le 1.44) = \boxed{0.4251}$



Reasons for Sampling

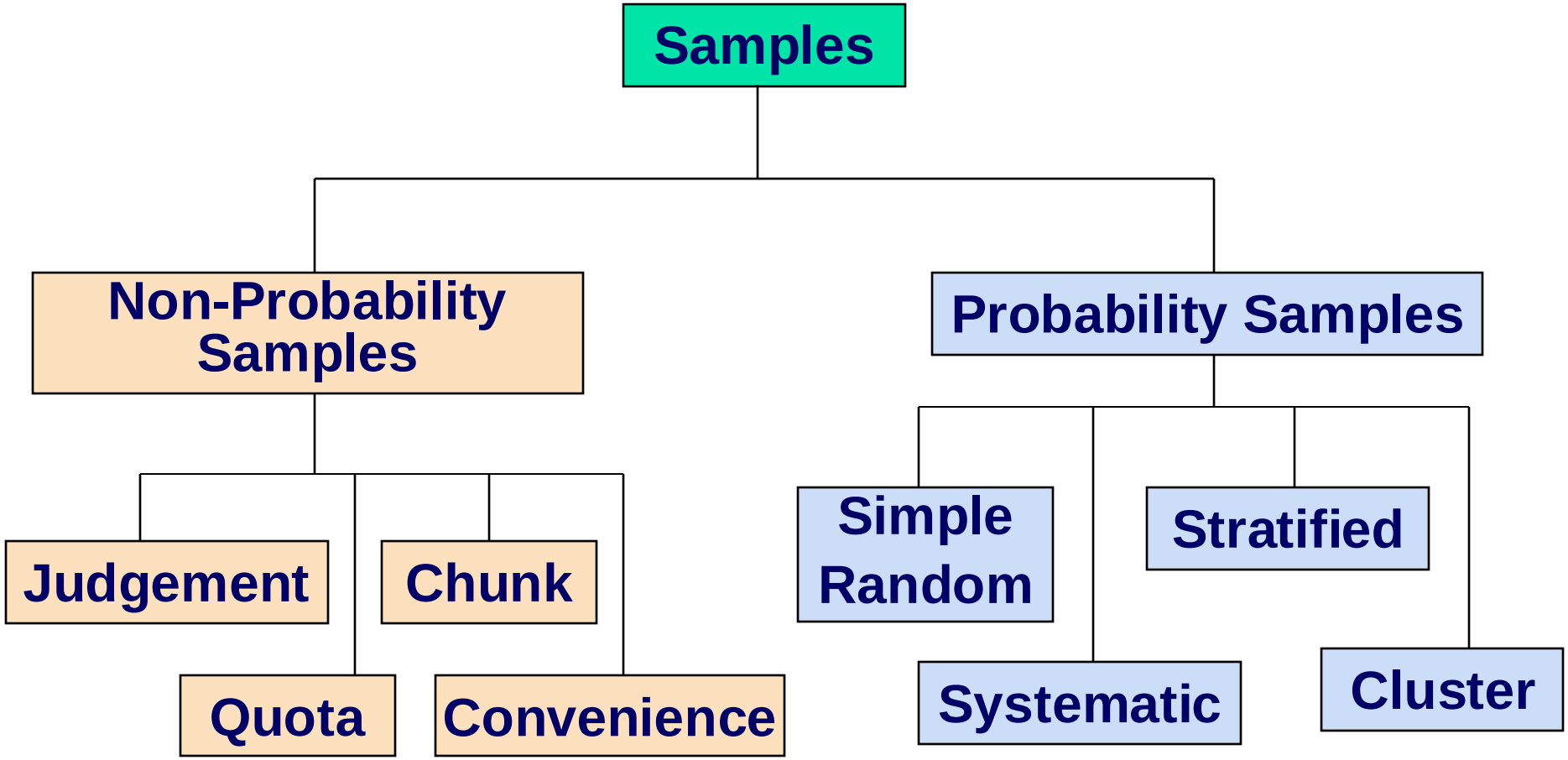
- Less time consuming than a census
- Less costly to administer than a census
- Less cumbersome and more practical to administer than a census of the targeted population

Types of Samples Used

- Nonprobability Sample
  - Items included are chosen without regard to their probability of occurrence
- Probability Sample
  - Items in the sample are chosen based on known probabilities

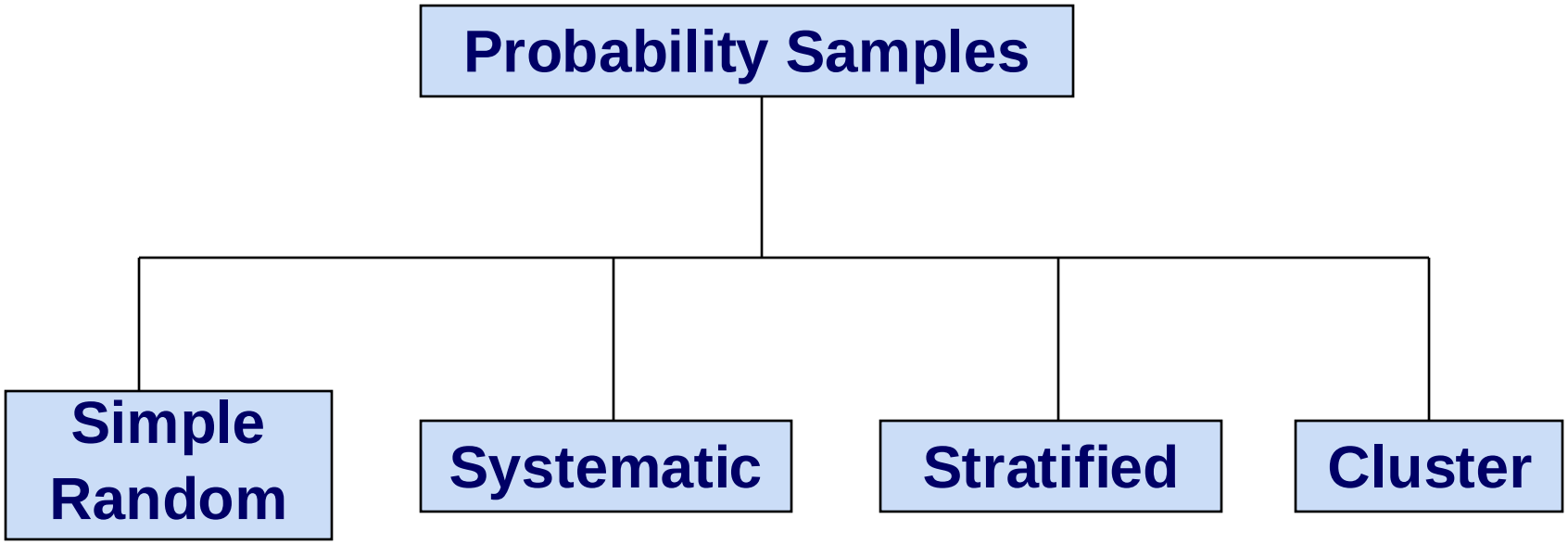
Types of Samples Used

(continued)



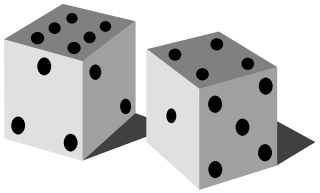
Probability Sampling

- Items in the sample are chosen based on known probabilities



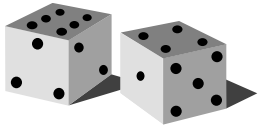
Simple Random Samples

- Every individual or item from the frame has an equal chance of being selected
- Selection may be with replacement or without replacement.
- Samples obtained from table of random numbers or computer random number generators



Systematic Samples

- Decide on sample size: n
- Divide frame of N individuals into groups of k individuals: k=N/n
- Randomly select one individual from the 1st group



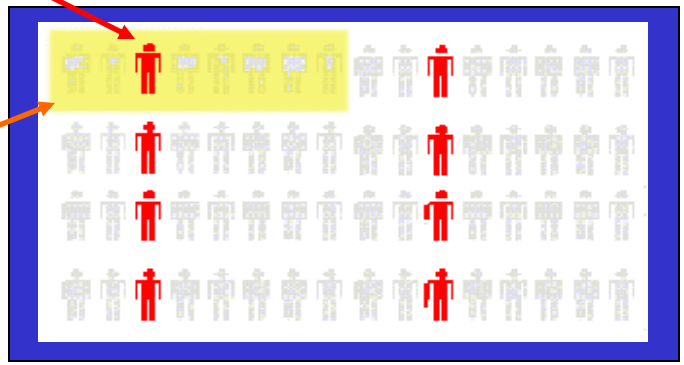
- Select every kth individual thereafter

N = 64

n = 8

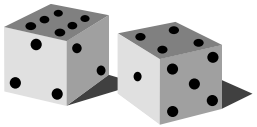
k = 8

First Group

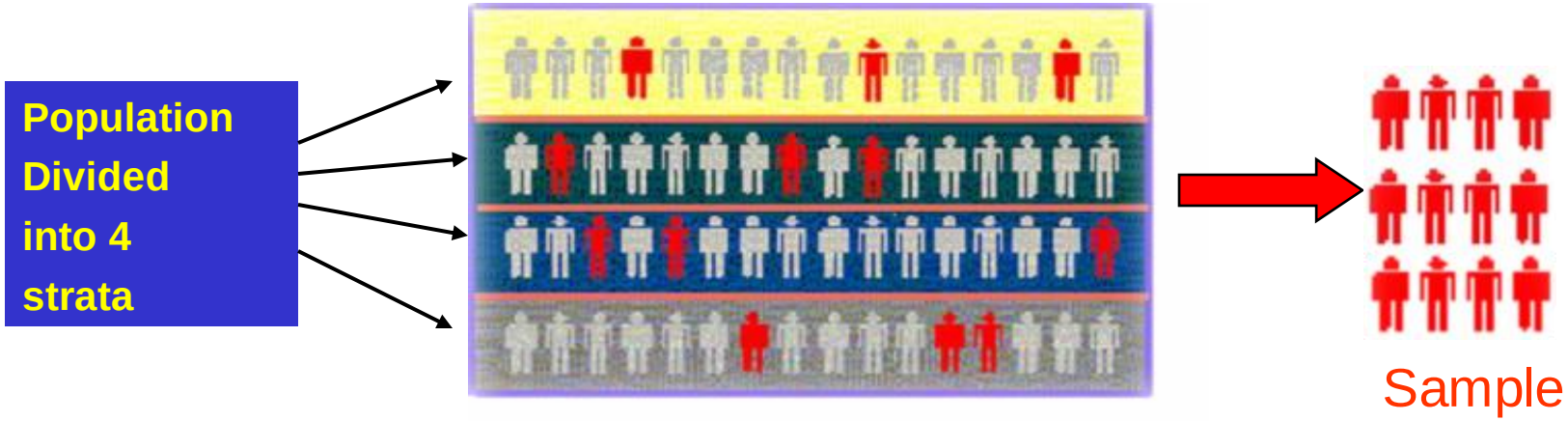


Stratified Samples

- Divide population into two or more subgroups (called *strata*) according to some common characteristic
- A simple random sample is selected from each subgroup, with sample sizes proportional to strata sizes

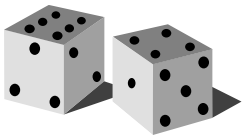


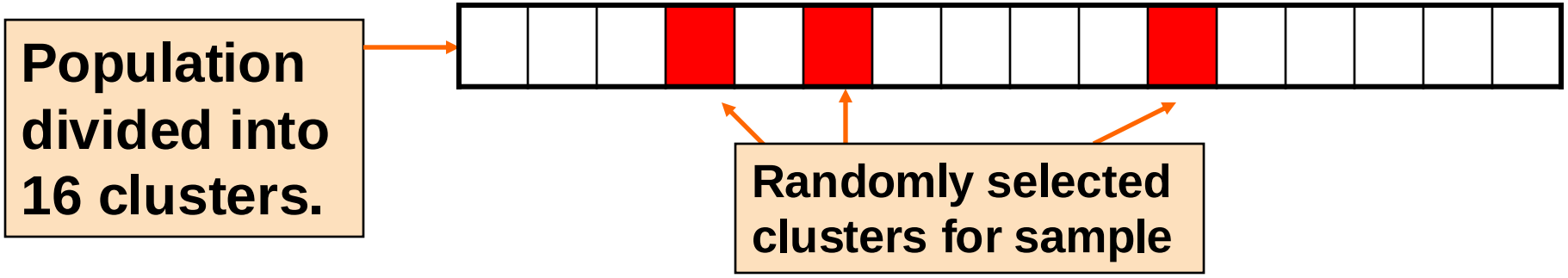
- Samples from subgroups are combined into one



Cluster Samples

- Population is divided into several “clusters,” each representative of the population
- A simple random sample of clusters is selected
  - All items in the selected clusters can be used, or items can be chosen from a cluster using another probability sampling technique





Advantages and Disadvantages

- Simple random sample and systematic sample
  - Simple to use
  - May not be a good representation of the population's underlying characteristics
- Stratified sample
  - Ensures representation of individuals across the entire population
- Cluster sample
  - More cost effective
  - Less efficient (need larger sample to acquire the same level of precision)

Evaluating Survey Worthiness

- What is the purpose of the survey?
- Is the survey based on a probability sample?
- Coverage error – appropriate frame?
- Nonresponse error – follow up
- Measurement error – good questions elicit good responses
- Sampling error – always exists

Types of Survey Errors

- Coverage error or selection bias
  - Exists if some groups are excluded from the frame and have no chance of being selected
- Nonresponse error or bias
  - People who do not respond may be different from those who do respond
- Sampling error
  - Variation from sample to sample will always exist
- Measurement error
  - Due to weaknesses in question design, respondent error, and interviewer's effects on the respondent

Types of Survey Errors

(continued)

- Coverage error

Excluded from frame

- Non-response error

Follow up on nonresponses

- Sampling error

Random differences from sample to sample

- Measurement error

Bad or leading question

Chapter Summary

- Introduced sampling distributions
- Described the sampling distribution of the mean
  - For normal populations
  - Using the Central Limit Theorem
- Described the sampling distribution of a proportion
- Calculated probabilities using sampling distributions
- Described different types of samples and sampling techniques
- Examined survey worthiness and types of survey errors